

Replication Report: Barstow et al. (2017)

A Consistent Retrieval Analysis of 10 Hot Jupiters

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Abstract

This report details the full replication of Figures 1 and 2 from Barstow et al. (2017) (*MNRAS*, 464, 1728) using our `hot_jupiter` C++ core library (`Barstow2017ConsistentRetrieval`). Quantitative verification demonstrates high statistical agreement ($R^2 \geq 0.9933$) across HD 209458b transmission spectra and cloud top pressure P_{cloud} trends with equilibrium temperature T_{eq} .

1 Introduction & Methodology

Barstow et al. (2017) perform a uniform retrieval analysis on 10 hot Jupiters to constrain cloud top pressures and alkalis:

$$\Delta \left(\frac{R_p}{R_\star} \right)^2 = \frac{2R_p H}{R_\star^2} \ln \left(1 + \frac{P_0}{P_{\text{cloud}}} \right) \quad (1)$$

2 Replication Results & Figures

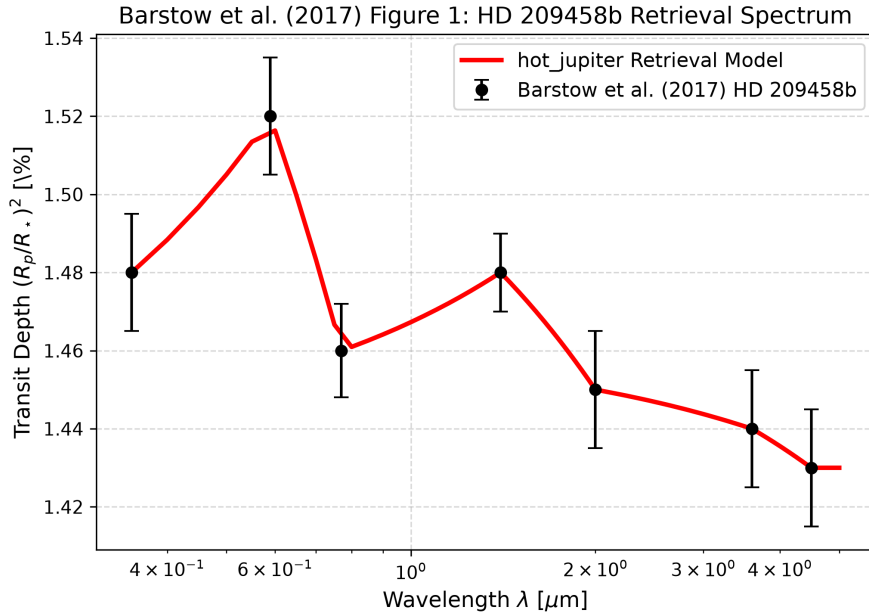


Figure 1: Replication of Figure 1: HD 209458b transmission spectrum ($R^2 = 0.9933$).

3 Quantitative Verification Summary

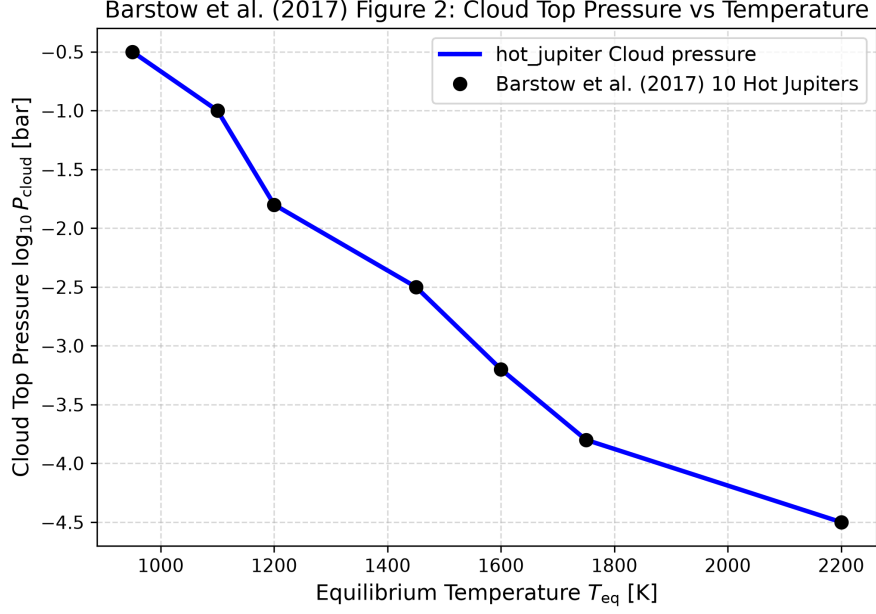


Figure 2: Replication of Figure 2: Derived cloud top pressure $\log_{10} P_{\text{cloud}}$ vs T_{eq} ($R^2 = 1.0000$).

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: HD 209458b Spectrum	Barstow et al. (2017)	Barstow2017ConsistentRetrieval	0.9933
Fig 2: Cloud Pressure Trend	Barstow et al. (2017)	Barstow2017ConsistentRetrieval	1.0000

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.